

UFF-SLFA v1.0.0

A Preregistered Sky-Lattice Falsification Audit

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7 August 2026

Abstract

Claims that astronomical catalogue anomalies cluster around a rigid celestial lattice are straightforward to state but unusually easy to test circularly. A query can select high-error records inside proposed node windows and then report that the selected records occur at the nodes. UFF-SLFA formalizes a neutral counter-framework that prevents this failure mode. A machine-readable contract freezes node coordinates, cap radius, anomaly rule, data split, null model, multiplicity correction, effect threshold, and decision rule before confirmatory analysis. The implementation compares anomaly rates inside and outside the frozen caps under footprint-aware right-ascension shifts, proper SO(3) rotations, or stratified label permutations. Every geometric null transform is checked for proper-rotation and lattice-invariance residuals, following the reproducibility discipline developed in QSOL TFT. Runs produce hashed recipe, observation, node-table, and manifest artifacts and support independent numerical replay. The result applies only to the exact catalogue-level association claim. It neither proves a physical ontology when positive nor claims that no later modified theory can be written when negative.

1. Problem

A celestial-node claim has three separable layers: a mathematical configuration of fixed sky coordinates; an empirical claim that a declared anomaly occurs unusually often near those coordinates; and a physical interpretation of any association. Only the second layer is tested here. Mixing the layers allows a valid geometry to protect an invalid empirical test, or an interesting catalogue association to be promoted directly into a field theory.

The central anti-circularity rule is simple: catalogue selection must be independent of the proposed node masks. A node-targeted sample cannot validate concentration at the nodes. Confirmatory evaluation also requires an independent catalogue or an untouched holdout split.

2. Frozen contract

The contract records ICRS coordinates, spherical-cap radius, anomaly threshold, catalogue hashes, holdout and weighting columns, null model, random seed, permutation count, family-wise alpha, minimum effect, and required supported-node count. The canonical contract hash changes if any field changes.

3. Selection-aware statistic

For anomaly indicator Y_i , positive survey weight w_i , and membership M_i in the union of node caps, the preregistered statistic is the weighted anomaly-rate contrast between the cap union and its complement.

$$\begin{aligned} p_{in} &= \text{sum}(w_i Y_i M_i) / \text{sum}(w_i M_i) \\ p_{out} &= \text{sum}(w_i Y_i (1 - M_i)) / \text{sum}(w_i (1 - M_i)) \\ T &= p_{in} - p_{out} \end{aligned}$$

The same contrast is calculated for every node. A Haldane-Anscombe corrected odds ratio is reported only as a diagnostic and does not replace the frozen statistic.

4. Selection-aware null models

Null	Preserved structure	Appropriate use
RA shift	Declinations, cap sizes, mutual geometry	Declination-dependent exposure; RA exchangeability
SO(3)	All angular separations and complete geometry	Full-sky or explicitly modelled selection
Stratified labels	Coordinates and anomaly burden within strata	Survey, depth, crowding, exposure, quality strata

For B null replicates, the one-sided empirical p-value is computed with a plus-one correction so a finite permutation experiment never reports $p = 0$.

$$p = (1 + \text{count}(T_b \geq T_{\text{obs}})) / (B + 1)$$

Node-wise p-values are corrected using Holm's step-down family-wise procedure. A node survives only when its adjusted p-value and effect size both satisfy the frozen criteria.

5. TFT invariance bridge

A null rotation must move the lattice without changing it. For every geometric replicate the implementation checks the following proper-rotation conditions and then verifies invariance of the node Gram matrix and all pairwise angular separations.

$$R^T R = I, \det(R) = +1$$

The maximum orthogonality, determinant, Gram-matrix, and pairwise-angle residuals are stored in the observations. This validates the transformation used by the audit; it is not evidence that the proposed celestial structure is physically real.

6. Decision and evidence bundle

A positive decision requires the global test to meet alpha and minimum effect and at least the preregistered number of nodes to survive Holm correction. Every node remains visible, including failures and sparse untestable caps.

Artifact	Role
recipe.json	Frozen contract, input hashes, row counts, replay boundary
observations.json	Global decision, node-count result, invariant residuals, claim boundary
nodes.csv	Every node, effect, raw p-value, Holm p-value, pass/fail
manifest.json	Byte sizes, SHA-256 hashes, runtime metadata, result

Verification separates artifact integrity from numerical replay. A bundle can be internally consistent and still be based on a poor scientific model. Hashes and replay establish computational consistency, not physical truth.

7. Interpretation boundary

UFF-SLFA refuses two symmetrical overclaims. A positive association does not prove a quantum-vacuum crystal, standing wave, topological defect, or other ontology. A negative result does not establish that an author could never modify a theory. It rejects the exact empirical claim encoded in the frozen contract.

Operational database outcomes also require disciplined language. Missing values, high uncertainty, excess-noise fields, morphology-fit diagnostics, and query timeouts are not physical objects or supercomputer crashes by default. Broadband colour differences are not spectral resonances without a bandpass-aware spectral model.

8. Conclusion

The framework replaces personality contests and self-adjudicated bounties with an executable claim contract. The decisive question is no longer whether a critic can prove an entire research programme permanently unpatchable. It is whether a specific, frozen prediction survives an independent, selection-aware, replayable test.

Software and related work

Reference implementation: QSOLKCB/UFF, modules uff.sky_audit and uff.sky_geometry. The rotation-invariance and artifact-boundary design is conceptually bridged from QSOLKCB/TFT: Reproducible Tensor Invariance and Quadrature Sonification. UFF-SLFA reimplements the required mechanisms under the UFF Apache-2.0 codebase rather than copying TFT's CC BY 4.0 implementation.

References

- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65-70.
- Phipson, B., and Smyth, G. K. (2010). Permutation p-values should never be zero. *Bioinformatics*, 26(2), 250-251.